

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer. Possesses a Master of Science in Computer Science with expertise in building scalable microservices and distributed data pipelines, leveraging technologies like Java, Python, Spark, SQL, AWS, and Azure. Proven ability in CI/CD automation, large-scale data platforms, and privacy-preserving AI applications, with experience in messaging queues like SQS and Kafka, and event streaming with Kinesis.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Engineered and automated CI/CD pipelines with Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring, reducing issue resolution time by 20 percent.
- Spearheaded a major EDX platform upgrade, streamlining application architecture and dependency management for 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, improving system performance by 90 percent through SQL query optimization and database tuning.
- Implemented CI/CD pipelines utilizing Jenkins and GitLab with automated testing and quality gates, ensuring robust deployment processes.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering team skill development.
- Integrated messaging services like SQS to enhance inter-service communication and ensure reliable data transfer.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage.
- Supported release planning and sprint execution using JIRA within Agile development teams, ensuring timely project delivery.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, AWS, Terraform) January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real-time, integrating with AWS SQS for message queuing.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization using Tableau.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and deployed Kinesis for real-time data streaming.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight.

Privacy-Preserving AI Email Client (Capstone Project, T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, utilizing Hugging Face libraries.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, demonstrating proficiency in privacy-preserving AI.

SKILLS

- Programming Languages: Java, Python, SQL.
- Frameworks and Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda, SQS, Kinesis), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming,

CI/ CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, Event Streaming.

- Web Services: REST API, SOAP API.